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Development of a Fast Parallel Hydraulic Fracture Simulator for Shale Reservoirs with Complex Natural Fractures

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Abstract

Hydraulic fracturing is a cornerstone technology in the development of unconventional oil and gas reservoirs. However, accurately modeling fracture propagation in geologically complex formations—such as those characterized by stress anisotropy, bedding interfaces, and natural fracture networks—remains computationally challenging. This study presents the development and application of a fast, parallel hydraulic fracture simulator designed to address these challenges by combining the Displacement Discontinuity Method (DDM) and Newton-Raphson iteration within an OpenMP-accelerated architecture. The simulator models both fracture propagation and fluid flow using dynamically growing matrix systems, which reflect evolving fracture complexity. A traditional serial LU decomposition solver is first implemented and benchmarked against a novel parallel solver using OpenMP-based multifrontal techniques, symbolic factorization, and optimized BLAS routines. The performance of the two solvers is evaluated using a field-realistic case from the Sichuan Basin, China—a region known for dense natural fracture networks and heterogeneous geomechanical conditions. Inputs include layer-specific properties such as Young's modulus, Poisson's ratio, and in-situ stresses, as well as a fully integrated 3D Discrete Fracture Network (DFN). Simulation results confirm that the parallel solver achieves a 10× speedup compared to the serial solver, without loss of accuracy in pressure propagation or fracture geometry. The simulator is further scaled to a 10-well cube development configuration spanning four vertical layers, demonstrating its capability to handle large, multi-well models in under an hour. This work offers a scalable, accurate, and computationally efficient solution for simulating complex fracture networks, enabling rapid, high-fidelity modeling for field development optimization. The platform lays the groundwork for future integration with AI-based optimization and real-time digital twin applications.

Introduction

Hydraulic fracturing has revolutionized the development of unconventional oil and gas reservoirs by making it economically viable to extract hydrocarbons from ultra-low permeability formations such as shale. These reservoirs are often characterized by intricate geological features, including multi-scale heterogeneity, high stress anisotropy, bedding interfaces, and networks of natural fractures. In such environments, fracture

propagation behavior becomes highly nonlinear and unpredictable, making accurate modeling not just beneficial—but essential—for effective well stimulation design and optimization.

To understand and control fracture growth under these complex subsurface conditions, numerical simulation has become a cornerstone of modern reservoir engineering. Advanced simulators enable engineers to predict fracture geometry, assess stage performance, and optimize treatment schedules based on formation properties. However, these simulations are computationally demanding. The challenge becomes especially pronounced when modeling three-dimensional rock discontinuities and fracture interactions with pre-existing natural fracture networks, where the coupling of rock mechanics and fluid flow dramatically increases the computational burden.

Among the various numerical approaches used for hydraulic fracture modeling, the Displacement Discontinuity Method (DDM) has gained prominence for its ability to efficiently represent complex fracture geometries using boundary element formulations. Unlike volume-based methods such as finite elements, DDM focuses computational resources along fracture surfaces, offering significant savings in memory and mesh generation. When coupled with the Newton-Raphson iterative method, which is commonly used to solve the nonlinear equations governing fluid pressure distribution and fracture propagation, the simulation framework becomes powerful yet computationally intensive. This coupling generates large matrix systems: dense matrices from rock deformation calculations and sparse matrices from fluid flow dynamics. As fractures propagate, these matrices grow rapidly in size, creating a major performance bottleneck—especially in multi-stage horizontal well simulations with hundreds of fracture clusters.

Recent advances in parallel computing offer a promising solution to these performance challenges. By distributing the workload across multiple CPU cores, parallel solvers can significantly reduce simulation time without compromising accuracy. In particular, OpenMP-based parallelization techniques are relatively easy to implement and highly effective on modern multi-core processors, making them ideal for integration into fracture simulation platforms. Despite these advantages, the performance of parallel solvers compared to traditional single-threaded solvers (such as serial LU decomposition) has not been extensively quantified in the context of hydraulic fracture simulations involving complex natural fracture networks.

To fill this gap, this study presents the development and benchmarking of a fast parallel hydraulic fracture simulator tailored for shale reservoirs with intricate geological features. The simulator integrates DDM and Newton-Raphson algorithms within a parallelized computational architecture using OpenMP. We conduct a detailed comparison between the parallel solver and a conventional single-core LU solver, evaluating both accuracy and runtime efficiency. A realistic field case is used to demonstrate the simulator's capabilities and to quantify the performance gains achievable through parallel computing.

By delivering a systematic performance evaluation, this paper aims to provide practical guidance for reservoir engineers and simulator developers on how to leverage parallelism in hydraulic fracture modeling. Our results contribute to ongoing efforts to accelerate complex geomechanical simulations, thereby enabling faster, more accurate decision-making in unconventional reservoir development. Ultimately, the proposed simulator advances the state-of-the-art in real-time and high-resolution fracture modeling, laying the groundwork for future integration with AI-driven optimization and digital twin platforms.

Methodology

The hydraulic fracture simulator employed in this study is based on the Displacement Discontinuity Method (DDM), a robust boundary element technique that is widely used for modeling fracture mechanics and fluid transport in fractured media (Wu and Olson, 2015a; Wu and Olson, 2015b). DDM allows for the simulation of both fracture propagation and fluid flow within evolving hydraulic fracture networks, making it highly suitable for capturing the complex interactions inherent in tight reservoirs.

A key feature of the simulator is its ability to explicitly model interactions between hydraulic fractures (HFs) and natural fractures (NFs). These interactions significantly influence fracture geometry, complexity,

and propagation paths—factors that ultimately affect reservoir stimulation efficiency and production performance. An example of such interaction modeling is illustrated in Figure 1, where multiple propagating hydraulic fractures intersect with a pre-existing natural fracture network. This capability enables the simulator to provide a more realistic and field-representative depiction of fracture growth, stress shadowing, and connectivity.

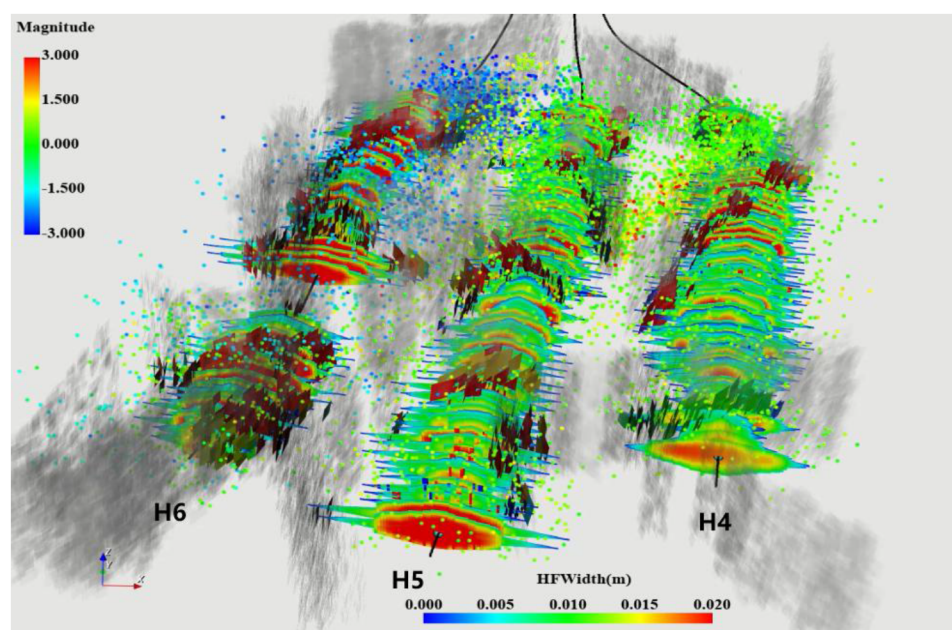


Figure 1—A hydraulic fracture simulator for simulating hydraulic fracture propagation in a shale reservoir with complex natural fracture distribution.

The simulator resolves two tightly coupled physical systems at each simulation time step: Rock Deformation System: Modeled using a dense matrix, this system governs the mechanical behavior of the rock, including stress redistribution and fracture propagation due to displacement discontinuities. Fluid Flow System: Captured using a sparse matrix, this system solves the pressure distribution and fluid flow within the fracture network and between fractures and the surrounding formation.

As the hydraulic fractures propagate over time, both matrices grow dynamically, reflecting the expanding complexity of the fracture network. The matrix size typically starts from $[2 \times 2]$ for early stages and can increase to $[5000 \times 5000]$ or larger for complex multi-fracture systems. This poses substantial computational demands, especially under serial processing schemes.

Serial Solver Framework

Initially, a serial direct solver based on LU factorization was implemented for both the dense and sparse systems. The procedure follows these steps:

Step 1: LU Decomposition: Decompose the coefficient matrix A into lower (L) and upper (U) triangular matrices such that $A = LU$.

Step 2: Forward substitution: let $y = Ux$, solve $Ly = b$ for intermediate vector y .

Step 3: Backward substitution: Solve $Ux = y$ to obtain the solution vector x .

While accurate, this method becomes inefficient as matrix dimensions increase.

Parallel Solver with OpenMP Acceleration

To address performance limitations and exploit modern multi-core hardware, a parallel direct solver was developed using OpenMP. The improved solver introduces pivoting strategies and multi-level parallelism for both fluid flow and rock deformation matrices. The system is solved in the form:

$$A = P_L L U P_R$$

where, P_L stands for row pivoting, which selects the largest element in a column to avoid division by small pivots, improving numerical stability. P_R stands for column pivoting, which reorders columns to place large elements on the diagonal or reduce fill-in for sparse matrices. Together, they ensure robustness and efficiency, especially for sparse or poorly conditioned matrices. L is a lower triangular matrix, U is an upper triangular matrix.

Once factored, the original $Ax = b$ system could be rewritten as $P_L L U P_R x = b$. The solution process includes 5 steps:

Step 1: Reordering (Symbolic Factorization): the new solver uses methods like METIS or AMD (Approximate Minimum Degree) for fill-reducing permutation, and computes the nonzero structure of L and U without numerical values.

Step 2: LU Decomposition: the new solver performs LU decomposition using parallel multifrontal or left-looking strategies. In multifrontal methods, independent frontal matrices are processed concurrently. During factorization, the new solver builds a tree of computation (elimination tree), allowing independent parts to be processed in parallel. Within each frontal matrix operation (BLAS Level 2/3), the new solver uses highly optimized routines (MKL or custom kernels) to accelerate.

Step 3: Forward substitution: let $z = P_R x$, $b' = P_L^{-1} b$, and $Uz = y$, solve $Ly = b'$ for y .

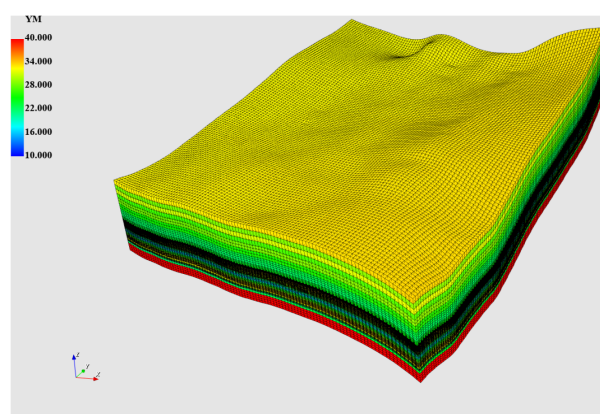
Step 4: Backward substitution: solve $Uz = y$ for z .

Step 5: Permute back: $x = P_R^{(-1)} z$.

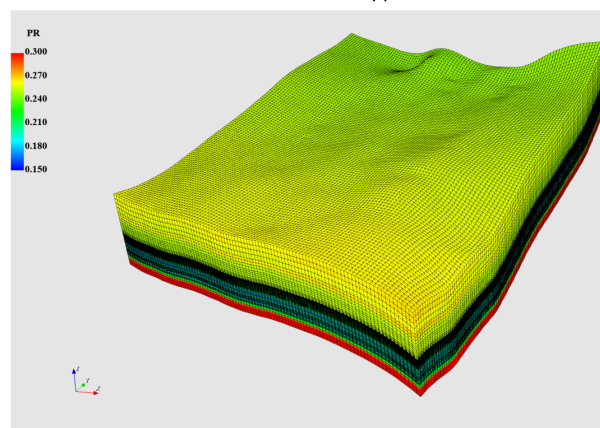
Comparison of Serial and Parallel Solvers in a Realistic Field Case

To evaluate the computational performance and scalability of the newly developed parallel solver, we conducted a detailed case study using a representative shale gas reservoir from the Sichuan Basin, China—a region characterized by complex geological conditions, including dense pre-existing natural fracture networks and high in-situ stress anisotropy (Yong et al., 2019; Zeng et al., 2021; Wu et al., 2023, 2024).

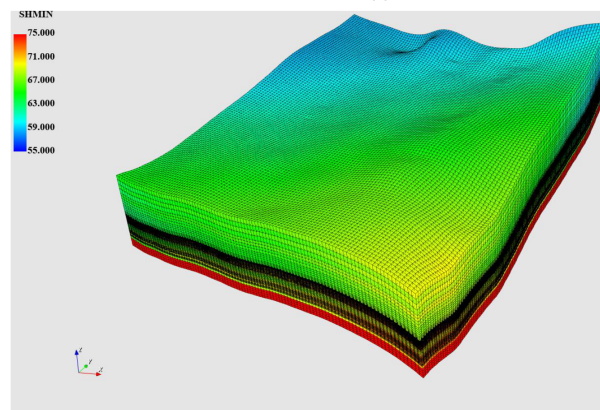
This case provides a rigorous test for the solver due to its highly heterogeneous geomechanical environment. The simulator incorporates spatially varying rock mechanics properties, including Young's modulus, Poisson's ratio, minimum horizontal stress, and maximum horizontal stress. These parameters are derived from an integrated geological and geomechanical modeling workflow and are illustrated in Figure 2. The horizontal well simulated in this study has a total lateral length of 1,550 meters, completed in a multi-stage hydraulic fracturing scheme.



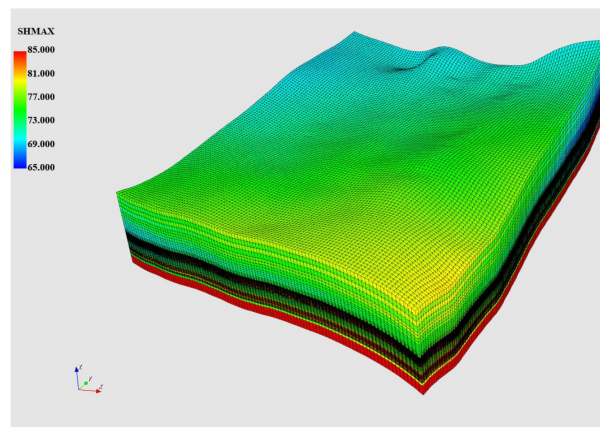
(a)



(b)



(c)



(d)

Figure 2—Spatial distribution of key geomechanical inputs: (a) Young's modulus; (b) Poisson's ratio; (c) minimum horizontal stress; (d) maximum horizontal stress.

These geomechanical inputs were directly incorporated into the hydraulic fracture simulator to ensure accurate modeling of rock deformation and fracture propagation behavior under complex in-situ conditions. The model accounted for spatial variability in mechanical properties and stress regimes, which critically affect fracture initiation, growth direction, branching, and containment. A summary of the key reservoir and mechanical properties used in the simulation is presented in [Table 1](#).

Table 1—Key model parameters for hydraulic fracture simulation

Model parameter	Value	Unit
Number of grid blocks ($x \times y \times z$)	$98 \times 116 \times 81$	-
Grid size ($x \times y \times z$)	$23 \times 23 \times 1.7$	m
Model size ($x \times y \times z$)	$2300 \times 2600 \times 140$	m
Reservoir depth	2080-2220	m
Young's modulus	18-49	MPa
Poisson's ratio	0.14-0.35	-
Minimum horizontal stress	57-75	MPa
Maximum horizontal stress	71-100	MPa

The model also incorporates a high-resolution 3D Discrete Fracture Network (DFN), formatted in FAB (Fracture Attribute Binary) format, enabling seamless integration into the simulator. As shown in [Figure 3](#), the DFN captures the spatial heterogeneity in fracture orientation, length, aperture, and density, which are prevalent in the Sichuan Basin. These natural fractures are essential to simulate accurately, as their interactions with induced fractures heavily influence fracture complexity, stimulated reservoir volume (SRV), and overall drainage efficiency. By integrating the DFN, the simulator can dynamically resolve how hydraulic fractures intersect with and activate natural fractures—leading to more complex and connected networks. This feature is essential for modeling multi-stage fracture designs and for understanding production responses, such as transient pressure behavior and long-term drainage performance.

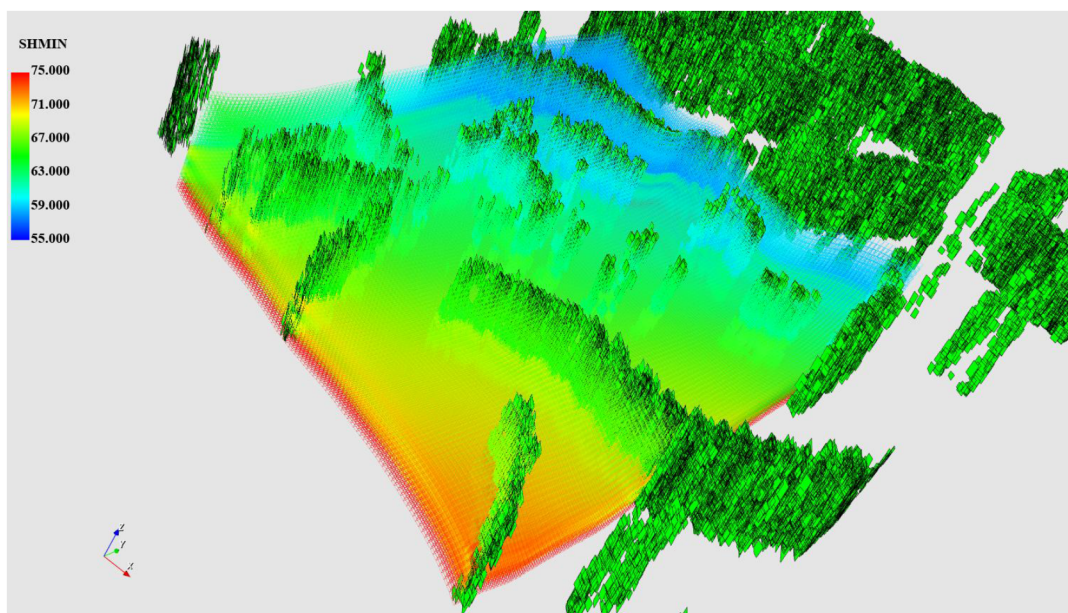
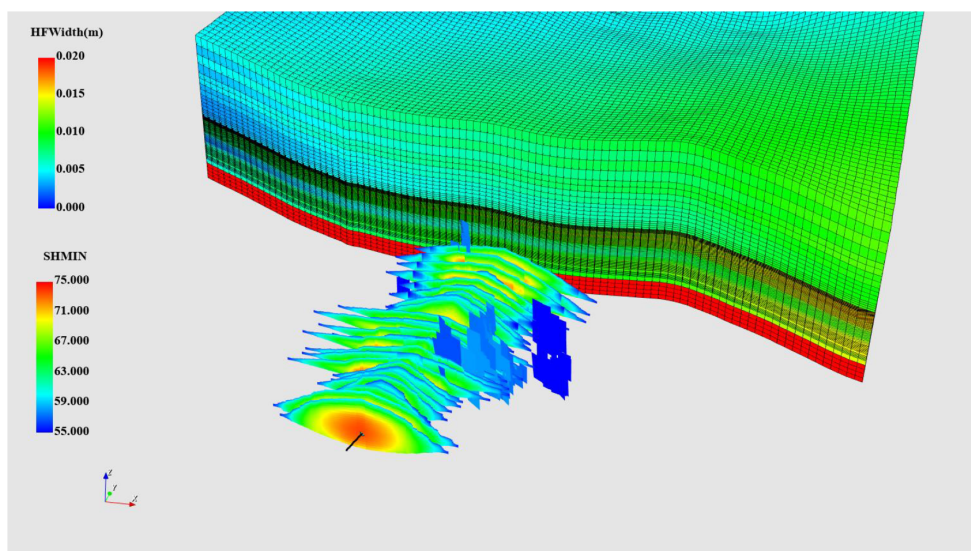
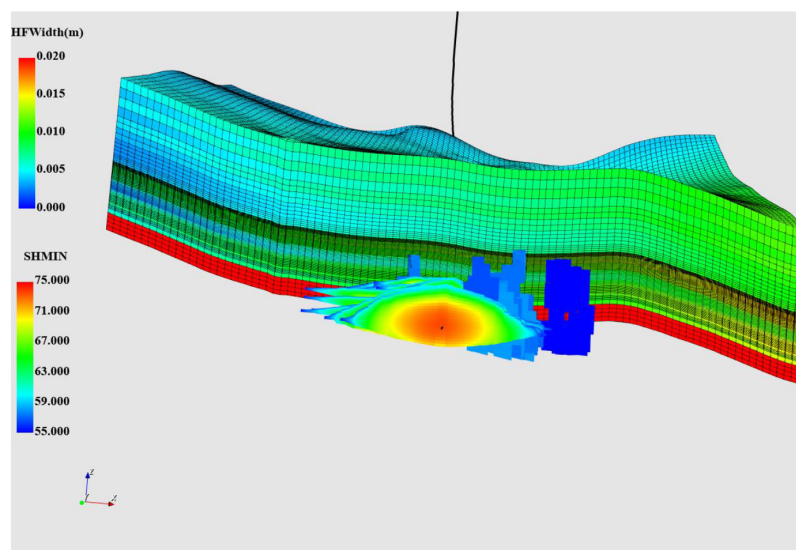


Figure 3—Three-dimensional DFN integrated into the geomechanical and hydraulic fracture model.

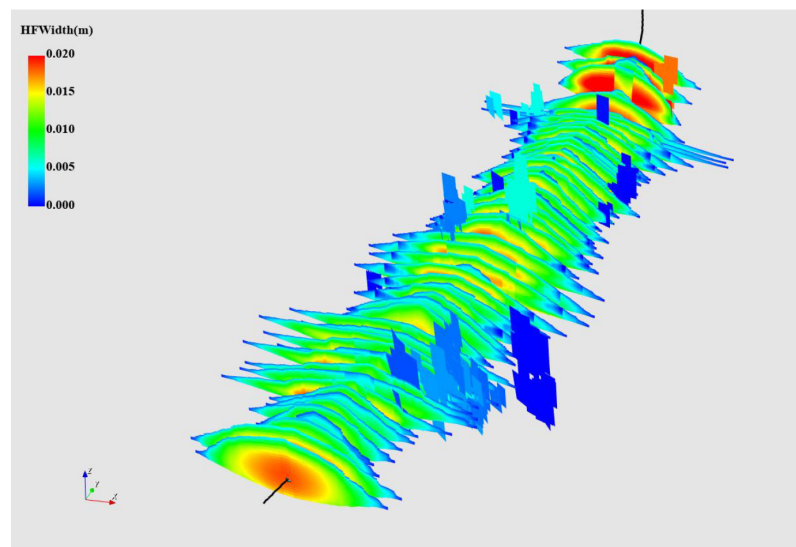
The resulting hydraulic fracture geometries, including both induced fractures and activated natural fractures, are shown in [Figure 4](#). These visualizations reveal the spatial extent, propagation direction, and connectivity of the fracture network, providing key insights into the effects of local stress perturbations, rock fabric heterogeneity, and pumping schedule variations on fracture growth. The case study was run using both the serial LU-based solver and the newly developed OpenMP-accelerated parallel solver. While both solvers produced identical results in terms of fracture geometry and pressure propagation, the computational efficiency was drastically improved with the parallel approach. Serial Solver Runtime: ~50 minutes per well simulation. Parallel Solver Runtime: ~5 minutes per well simulation. Observed Speedup: ~10× improvement. This speedup is significant for field-scale hydraulic fracturing campaigns, where dozens to hundreds of wells may need to be simulated iteratively for design optimization, history matching, or uncertainty quantification. The dramatic reduction in computation time without sacrificing accuracy highlights the scalability, robustness, and practical value of the parallel solver architecture in real-world applications. This comparative study in a geologically complex basin demonstrates that the parallel solver significantly enhances computational throughput while retaining modeling fidelity, thereby enabling rapid, high-resolution simulations of hydraulic fracture networks in structurally intricate reservoirs.



(a)



(b)



(c)

Figure 4—Simulated fracture network including natural fracture activation: (a) geomechanical background view; (b) side perspective view; (c) full-well geometry overview.

Field-Scale Application: Multi-Well Cube Development

To evaluate the scalability and performance of the simulator in a realistic field-scale setting, the model was expanded from a single-well configuration to a multi-well scenario involving 10 horizontal wells. These wells were strategically distributed across four vertical reservoir layers, reflecting the typical cube development strategy employed in tight shale plays. From bottom to top, the wells were grouped as follows: three wells in the lowest layer, followed by two wells in the second layer, three wells in the third layer, and two wells in the uppermost layer. This vertical stacking pattern is designed to maximize stimulated reservoir volume (SRV) while minimizing inter-well interference. All wells maintained a lateral length of 1,550 meters, consistent with the previous single-well case. The horizontal spacing between adjacent wells was approximately 500 meters, a spacing used in cube development to balance fracture complexity and production efficiency. The spatial configuration of the 10 wells is illustrated in Figure 5.

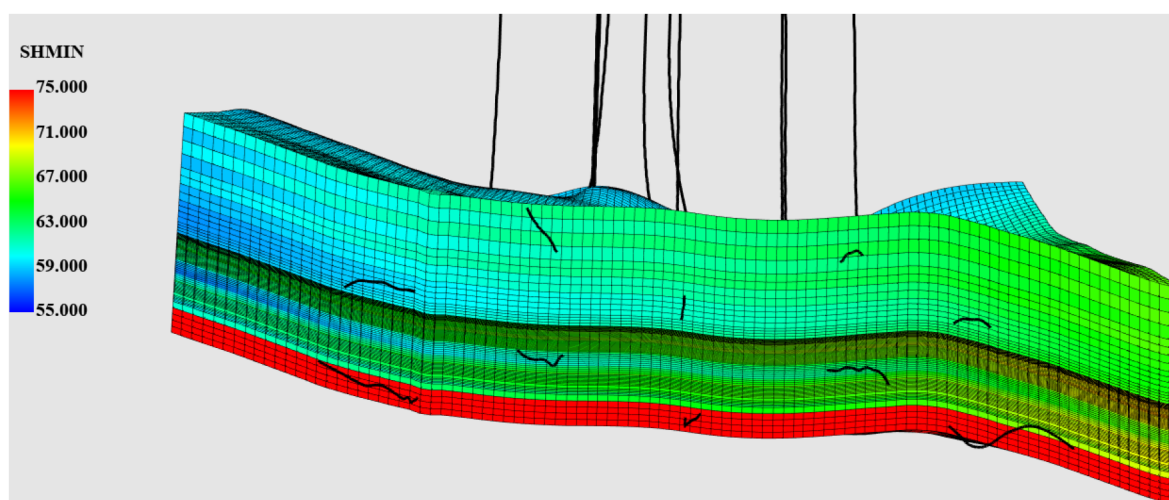


Figure 5—Spatial distribution of 10 horizontal wells across four stacked reservoir layers.

This multi-well case study is particularly important for simulating and optimizing cube shale development strategies, where the interactions between fractures in different layers can significantly influence well performance. Proper modeling of fracture height growth, inter-well stress interference, and stage sequencing is essential for maximizing productivity and minimizing wellbore damage or pressure communication. The simulated 3D fracture geometry for all 10 wells is presented in Figure 6. The results show the spatial extent and complexity of the hydraulic fractures, including the influence of reservoir heterogeneity and stress anisotropy on fracture propagation. The fracture planes from adjacent wells are clearly distinguished, enabling detailed analysis of potential interference zones and stimulation overlap across layers. Remarkably, the total simulation time for this 10-well case using the OpenMP-accelerated parallel solver was approximately 50 minutes, underscoring the solver's capability to handle large-scale, multi-well models with high computational efficiency. This level of performance is critical for operational workflows, where rapid simulation turnarounds are required for real-time decision-making, design iteration, and field development planning.

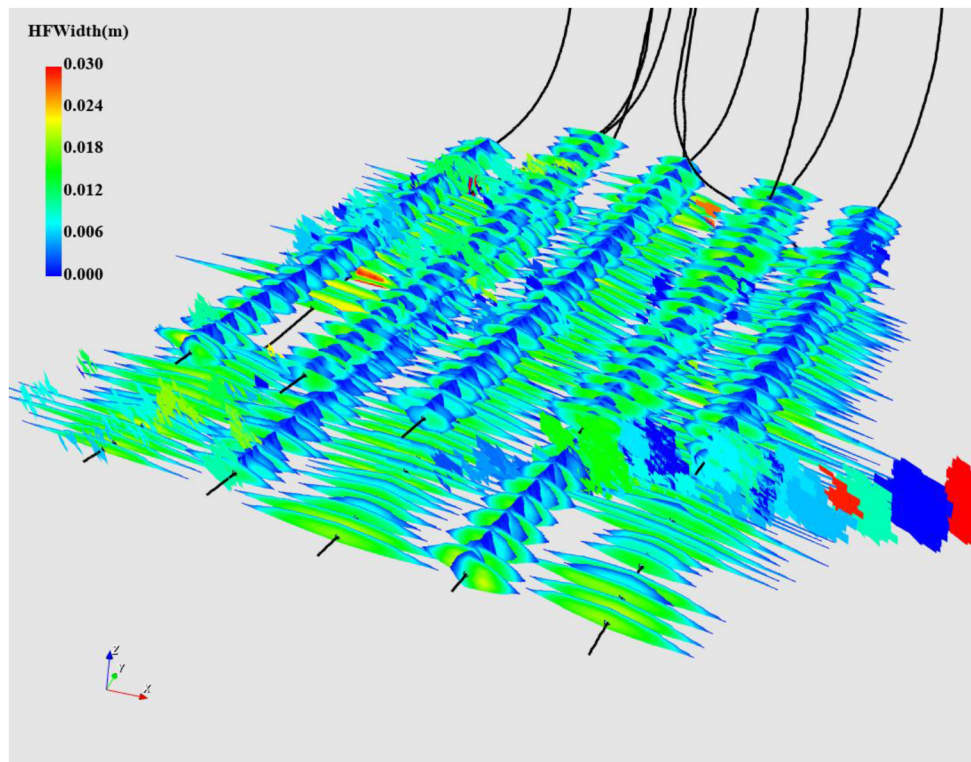


Figure 6—Simulated 3D fracture geometries of 10 horizontal wells in a cube development configuration.

Conclusions and Future Work

This study introduces a high-performance parallel hydraulic fracture simulator that addresses the computational limitations associated with modeling complex fracture growth in unconventional reservoirs. By integrating the DDM with Newton-Raphson iteration and parallelized via OpenMP, the simulator efficiently resolves coupled systems of rock deformation and fluid flow using dynamically evolving dense and sparse matrices. Key contributions and findings include:

1. **Accurate modeling of complex geology:** The simulator successfully incorporates spatially variable geomechanical inputs and a 3D Discrete Fracture Network (DFN), enabling realistic simulation of fracture–natural fracture interactions in heterogeneous shale formations.
2. **Robust solver performance:** The parallel solver achieves a 10× reduction in runtime relative to the serial LU solver while preserving simulation fidelity. This allows for rapid iteration and real-time application in operational settings.
3. **Field-scale validation:** The simulator was benchmarked against a real-world case from the Sichuan Basin and scaled to simulate 10 horizontally stacked wells in a cube development configuration. The full 10-well simulation completed in approximately 50 minutes, highlighting the solver's suitability for high-resolution, multi-well field development modeling.
4. **Scalability and practicality:** The modular solver architecture and parallel computing strategy make the simulator highly scalable and compatible with modern multi-core computing platforms. It provides a practical solution for reservoir engineers engaged in iterative design, history matching, and real-time decision support.

This work advances the current capabilities of hydraulic fracture simulation by enabling high-fidelity, high-efficiency modeling of complex subsurface behavior. Future directions will focus on coupling the simulator with geomechanical evolution models, machine learning–driven fracture design optimization, and integration into cloud-based digital twin platforms for real-time reservoir management. Through these

extensions, the simulator will play a pivotal role in accelerating unconventional resource development with enhanced accuracy and decision-making confidence.

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